Practical : 8

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Problem Statement: Greedy Algorithm (Fractional Knapsack) - CO1, CO2, CO5

Maximize Profit by Shipping Partial Orders (Fractional Knapsack) Problem Statement: You run a shipping company and need to load a truck with parcels of different weights and profits. The truck has a limited weight capacity. Write a program to choose parcels (even partially) to maximize profit using the Fractional Knapsack strategy.

Code

```
# Fractional Knapsack using Greedy Algorithm

n = int(input("Enter number of parcels: "))

items = []

for i in range(n):
    weight = float(input("Enter weight: "))
    profit = float(input("Enter profit: "))

    ratio = profit / weight

    items.append([weight, profit, ratio])

capacity = float(input("Enter truck capacity: "))

# Sort by profit/weight ratio in descending order
items.sort(key=lambda x: x[2], reverse=True) totalProfit = 0
```

```
print("\nSelected Parcels:")

for item in items:

    weight = item[0]
    profit = item[1]

    if capacity == 0:
        break

    if weight <= capacity:

        print("Full Parcel -> Weight:", weight,
              " Profit:", profit)

        capacity -= weight
        totalProfit += profit

    else:

        fraction = capacity / weight

        print("Partial Parcel -> Weight:", capacity,

              " Profit:", round(profit * fraction, 2))
```

```
        totalProfit += profit * fraction
        capacity = 0

print("\nMaximum Profit =", round(totalProfit, 2))
```

Output

Input

Enter number of parcels: 3

Enter weight: 10

Enter profit: 60

Enter weight: 20

Enter profit: 100

Enter weight: 30

Enter profit: 120 Enter

truck capacity: 50

Output:

Selected Parcels:

Full Parcel -> Weight: 10.0 Profit: 60.0

Full Parcel -> Weight: 20.0 Profit: 100.0

Partial Parcel -> Weight: 20.0 Profit: 80.0

Maximum Profit = 240.0